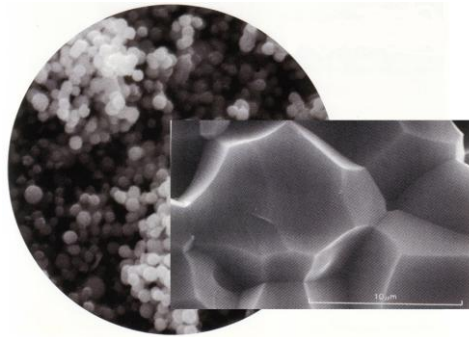




**Politecnico
di Torino**

Mechanical & Automotive
Engineering
A.Y. 2025-2026

TECHNOLOGY of METALLIC MATERIALS



Lecture 4 Microplasticity

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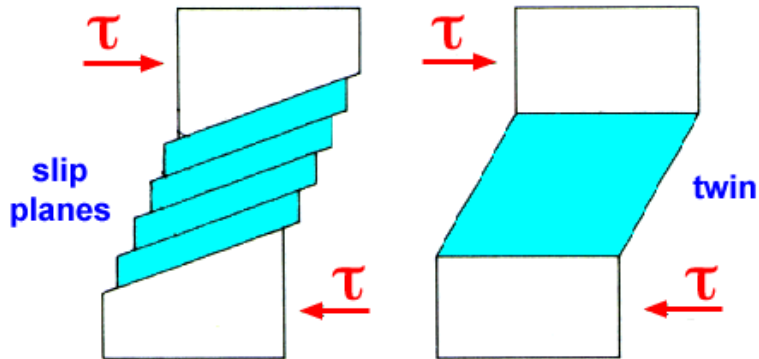
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Outline Lecture n.4

- Slip systems in single crystals and Schmid's law
- Grain boundary properties, types, energy (GBE)
- Plastic deformation of single crystal (review)
- Plastic deformation of polycrystals, the Taylor's law
- Microplasticity – yielding in polycrystals
- Relationship between uniform tensile deformation and strain hardening coefficient
- Strengthening of metals and alloys
- Cottrell's atmospheres
- Toughening of castings
- Sources of dislocations (Frank-Read source and GBs)
- Macroplasticity and strain hardening, work hardening coefficient
- Estimation of grain size in polycrystalline metals
- Luders' bands
- Plasticity in manufacturing

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Mechanisms of plastic deformation



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Slip systems in single crystals

Crystals exhibit special planes, called **slip planes**, over which dislocations move easily as compared to other planes. Such planes contain a high packing factor (atom density).

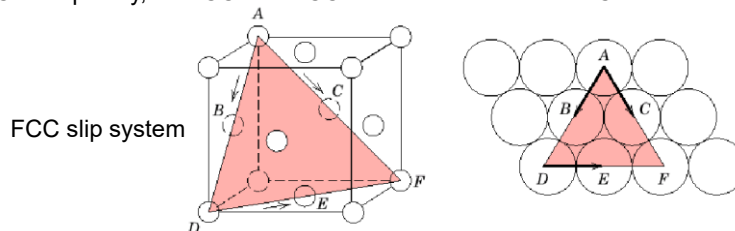
Dislocations preferentially slip over slip planes and along **highly dense directions**.

High dense plane families and high dense direction families (**slip direction**) identify a **slip system**.

BCC, FCC and HCP crystals have their own **characteristic slip systems**.

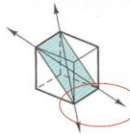
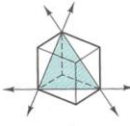
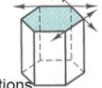
Because the distance between atoms is smaller than average one, the interplanar distance (to the normal direction) is larger than its average counterpart, allowing easier relative slip of these planes w.r.t. other planes.

BCC and FCC crystals possess more slip systems than HCP crystals, thereby having more freedom to let dislocations propagate; this confers to metals larger ductility. Consequently, $\Rightarrow$ FCC and BCC are more ductile than HCP.



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Slip systems in Metal crystals

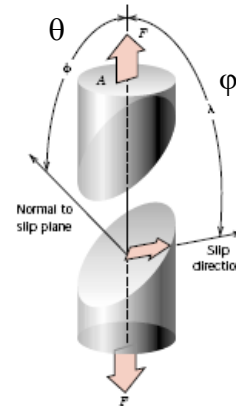
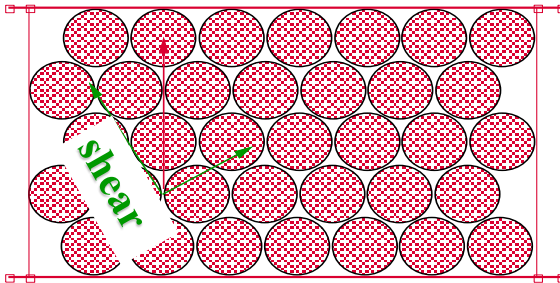
Crystal structure	Slip plane	Slip direction	Number of slip systems	Unit cell geometry	Examples
bcc	$\{110\}$ $\{112\}$ $\{123\}$	$\langle\bar{1}11\rangle$ // //	$6 \times 2 = 12$ $6 \times 2 = 12$ $6 \times 4 = 24$ In total: 48		α -Fe, Mo, W
fcc	$\{111\}$	$\langle\bar{1}10\rangle$	$4 \times 3 = 12$		Al, Cu, γ -Fe, Ni
hcp	$\{0001\}$	$\langle 11\bar{2}0 \rangle$	$1 \times 3 = 3$		Cd, Mg, α -Ti, Zn

Dislocations **can only move** on slip planes along slip directions.
The slip planes and directions are the closest packed in a lattice

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Plastic Deformation under normal stress

Any normal force F can be *resolved* into a **shear stress** component τ and a **normal stress** component σ stress over a slip plane. When the resolved shear stress τ exceeds a critical value, τ_{CRSS} , slip occurs in the “slip plane”



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The Schmid's Law

- For a dislocation to move over its slip systems, a shear stress acting over the slip plane and along the slip direction must be applied; this is typically determined by an applied normal force to the sample such as it happens in a tensile test
- A uniform tensile stress σ splits into a stress normal to the plane (sigma) and a **resolved shear stress** parallel to the plane and acting along a direction that lie at an angle θ to the tensile axis;
- this **resolved shear stress** is responsible of the dislocations motion, when its magnitude exceeds a critical level τ_{CRSS} ,
- **Critical Resolved Shear Stress τ_{CRSS}** is the stress at which macroscopic yield begins.
- Each material has its own τ_{CRSS} , thus it is a **material property**.
- It is closely related to yielding stress σ_y .
- *When a tensile stress is imposed, dislocations first move on the slip plane on which τ_{CRSS} is minimum.*

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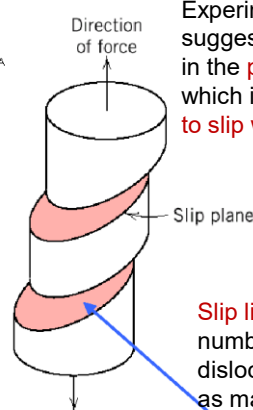
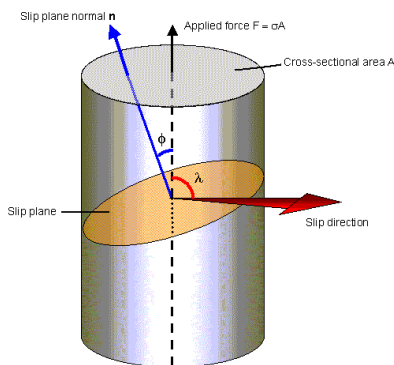
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Schmid's Law: tensile test of SC

Upon tensile test of a single crystal (SC) sample as shown in figure

$$F = \sigma A$$

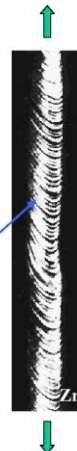
The **axial loading direction (F)** selected as **reference**
Two angles (ϕ , λ) define the **slip plane** and the **slip direction** respectively



Experimental evidence suggests that slip starts in the **primary slip system** which is the system **first to slip w.r.t. to imposed τ**

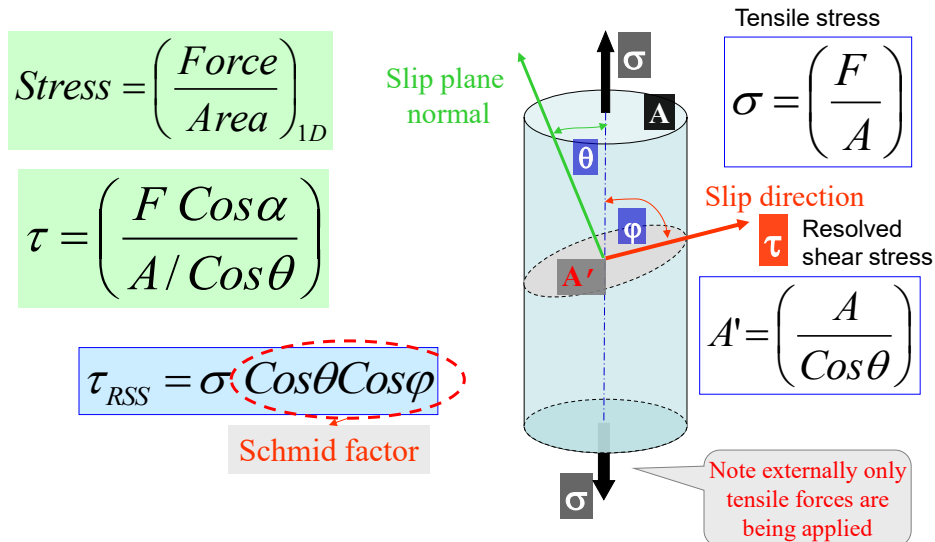
Slip bands are large number aggregates of dislocations

Slip lines are small number aggregates of dislocations and appear as markings over the sample surface



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Definitions in the Schmid's Law



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Critical Resolved Shear Stress

If the slip plane is perpendicular to the Tensile Stress
 $\theta = 90^\circ$, and $\phi = 0^\circ$

$$\cos \phi = 1 \text{ and } \cos \theta = 0 \Rightarrow \tau_{RSS} = 0$$

This means that even if the applied tensile stress is enormous, the resolved shear stress is simply zero. The dislocations can't move (slip can't occur).

For these extreme orientation the crystal ordinarily fractures rather than deforming plastically.

$\Rightarrow$ maximize efficiency of applied tensile stress with respect to angles orientation

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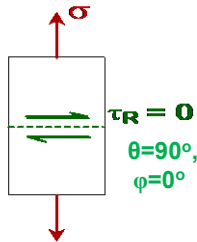
Critical Resolved Shear Stress

Condition for dislocation motion:
Crystal orientation with respect to imposed σ can make it easy or hard to move dislocation.

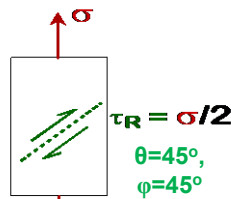
$$\tau_{RSS} > \tau_{CRSS}$$

typically
 10^{-4}G to 10^{-2}G

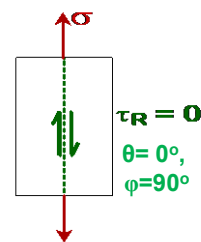
$$\tau_{RSS} = \sigma \cos\theta \cos\phi$$



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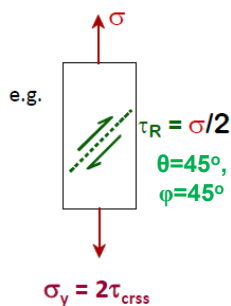
$$\tau_{RSS,max} = \frac{\sigma}{2}$$



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Schmid's law in Polycrystals (Taylor factor)?

Single crystal



polycrystalline



Center grain

$$\tau_R = \sigma/2$$

$\theta = 45^\circ$,
 $\phi = 45^\circ$ on average

For each grain,
 ϕ and θ are different in each grain

$$\sigma_y = \frac{\tau_{CRSS}}{(\cos\theta \cos\phi)_{max}}$$

Which, of either single crystal or polycrystal does require larger load to start slip?

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Critical Resolved Shear Stress

During slip:

interplanar distance stays constant

- the number of plain in the sample stays constant



$l \cos \phi = \text{const}$

With increasing length of the sample the angle between the normal to the plane and the load axis tends to 90°

$l \sin \lambda = \text{const}$

With increasing length the angle between the slip direction and the loading axis tends to 0°

slip initiates when

$$M \tau_{CRSS} = \sigma_{ys}$$

M = Taylor factor = 3.1
for each single crystal
in tensile/compression

tensile test single crystal oriented so that different values ($\cos \phi \sin \lambda$)_{max} are generated in most -favored slip system.

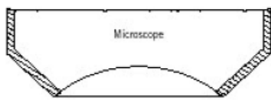
- *Measure σ_{vs} are different. But, σ_{vs}/m are invariant!*

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Grain Boundaries. Main features

- They are 2D defects, typical in polycrystalline materials
- They are produced by the relative rotation of two adjacent grains;
- The rotational angle can be:
 - Small (if $\theta \leq 15^\circ$) **small-angle GBs**
 - Large (if $\theta \geq 15^\circ$) **large-angle GBs**
- Whent the angle is small the GB can be viewed as constituted by a sequential array of edge dislocations
- In large-angle GBs, other than edge dislocations **there might simultaneously exist** other defects at the GB, therefore it is not correct anymore its description by only edge dislocations

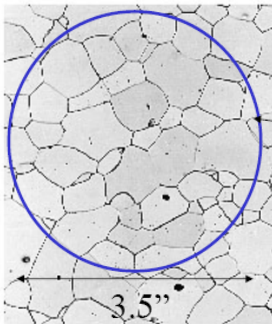
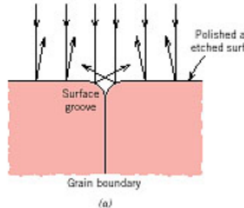
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Grain Size

MICROSTRUCTURE

ASTM (American Society for Testing And Materials)
grain size number n



W
100x

$$N = 2^{n-1}$$

100x with
area in²

N = no. of grains/in²

46 grains + 22 on circumference
= 46 + 22/2 = 57 grains

$$N = 57 / \pi (3.5/2)^2 = 6 \text{ grains/in}^2$$

$$6 = 2^{n-1}, n = 3.6$$

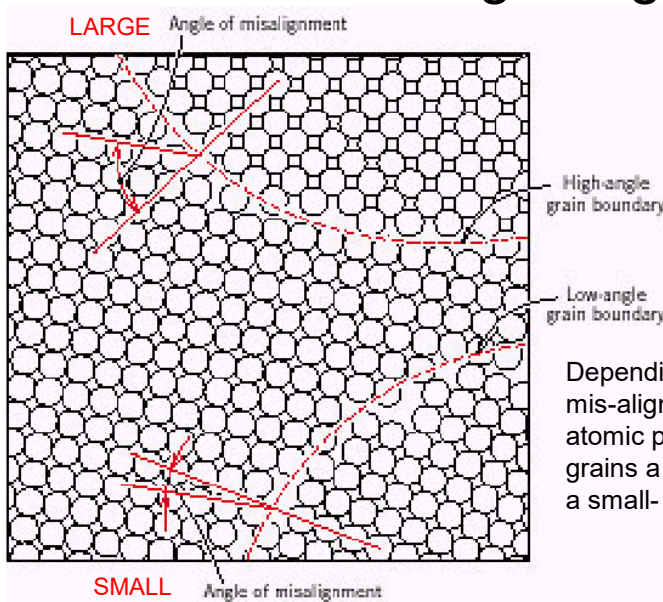
$n \nearrow$ $N \nearrow$

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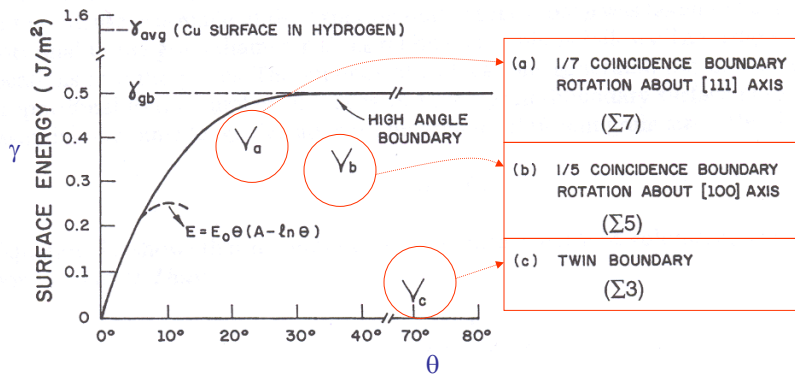
Small- and Large-angle GBs



Depending on the value of the mis-alignment angle between atomic planes of two adjacent grains a GB can be divided into a small- and large-angle GB

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Surface (GB) Energy as a function of misorientation angle θ (Cu case)



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Consequences of GBs

1. They are sources and sinks for dislocations
2. They contribute to mechanical strengthening and change the sliding system
3. They act as sinks for impurities dissolved in the alloy
4. They contribute to the formation of second phases – primary nucleation sites for the growth of second phases
5. They are material portions with no crystallinity

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Yielding in Polycrystalline Materials

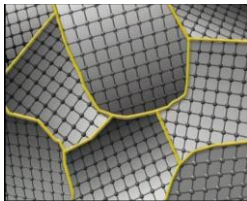
- **Polycrystals**, in the absence of texture, can be treated as **isotropic materials**
- Materials which are **ductile in single crystalline** form, **may not be ductile** in the polycrystalline form.
- For slip to occur in **polycrystalline** materials, at least **5 independent slip systems** must be active.
- **At higher temperatures** more slip systems may become active, hence polycrystalline materials which are brittle at low temperature, become ductile at high temperature.

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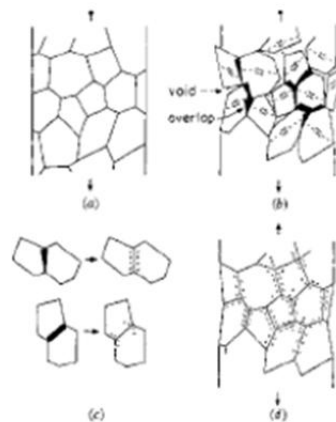
Yielding in Polycrystalline Materials



Slip planes & directions (θ , ϕ) of the same slip system change from one grain to another,



As grains are randomly oriented and HAGBs hinder the crossing of dislocations, polycrystal are much stronger than single crystal



Voids and overlap during plastic deformation are **NOT allowed**

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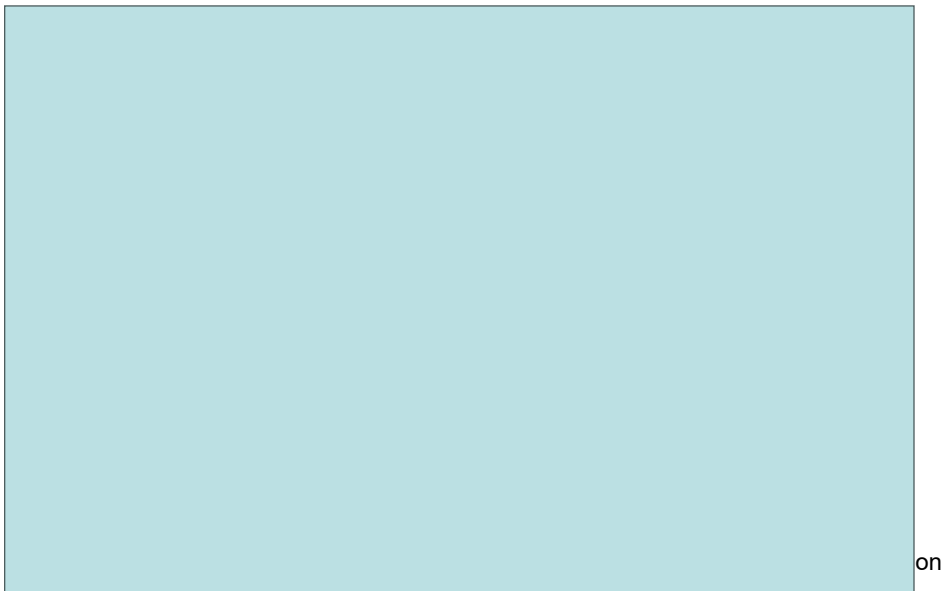
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Test your learning: what are the microstructure features ensuring an ideal strength-toughness combination? 1/2



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Test your learning: what are the microstructure features ensuring an ideal strength-toughness combination? 2/2



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Stacking fault Energy

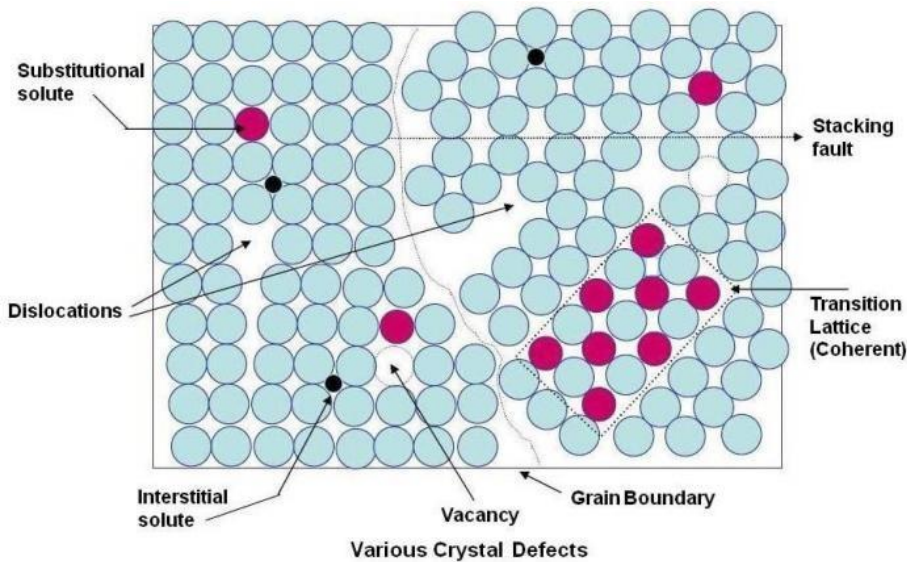
[35] L.E. Murr. Interfacial phenomena in metals and alloys. (1975) Addison-wesley, Reading

Table 1 SFE of common metals and alloys measured at room temperature [35]

Metal	γ_{SFE} (mJm ⁻²)	Metal	γ_{SFE} (mJm ⁻²)
Aluminium	166	Zinc	140
Copper	78	Magnesium	125
Silver	22	91Cu:9Si	5
Gold	45	Zirconium	240
Nickel	128	304 Stainless steel	21
Cobalt (FCC)	15	70Cu:30Zn	20

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Summary of lattice defects in Crystals



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SUMMARY OF DEFECT'S EFFECTS: ON PROPERTIES

Defect	type	Improved Materials Properties	Adversely affected Materials Properties
Point Defect	Vacancy $f(T)$	– Diffusivity – Color Centers – Ionic Conductivity	– Electron mobility – Carrier Lifetime
	Substitutional	– Conductivity (dopant) – Strength (hardness) – Characteristic T (like T_M)	– Conductivity (impurities) – Ductility – Characteristic T
	Interstitial	– Strength – Characteristic T – Electrical Properties	– Ductility – Characteristic T – Electrical Properties
Line Defect	Dislocation	– Ductility (Malleability) – Strength (at high density)	– Strength – Yield Stress – Optical Properties – Lasing Action
Planar Defect	Grain Boundaries	– Strength – Electrical Properties – Magnetic Properties	– Creep – Electrical Properties – Magnetic Properties

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Questions: dislocations

1. List the basic characteristics of a dislocation
2. What does cause dislocation motion
3. Find an analogy for a screw dislocation motion
4. When a dislocation does start/stop moving?
- 5 How the dislocation line is related with $\underline{b}$ vector in edge and screw dislocations
6. Is the crystal affected after crossing a dislocation?
7. Is dislocation motion continuous or discrete across the crystal?
8. Can a dislocation exist in polycrystalline ceramics?
9. If we want to observe a large dislocation density in a tensile tested sample in which region can it be found?

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Self-assessment Questions 1

1. Performance vs. properties: which manufacturing factors affect the mechanical response of a tensile sample vs. a true structural part
2. Performance vs. properties: compare the tensile strength of a material sample vs. that extracted from a structural part?
3. Performance vs. properties: following questions n.1 and n.2, guess similar answers assuming a complex geometry and large size of the structural part
4. Why the dimensional scale of microstructure does affect the material properties?
5. Which are the instruments allowing observation of microstructures at different dimensional scales?
6. How crystal-sensitive material properties are measured or diagnosed ? and list a few of them.
7. How the individual scales of microstructure do interact to each other?
8. Which lattice defects do affect mechanical properties ? List a few of them.
9. Does cold rolling affect the isotropy of properties? Which sample orientation is most penalized?
10. Are the properties of a cold rolled alloy isotropic or anisotropic after annealing?
11. How strength and toughness can affect the cold worked parts
12. Why the mechanical properties of polycrystalline materials are isotropic in spite of anisotropic properties of individual grains
13. How shall be designed the microstructure for best combination of strength and toughness

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Lecture 4

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Self-assessment Questions 2

11. Why of the exceptional combination of strength, toughness and sharp edge in Damascus swords: justify each performance with the received manufacturing process
12. How shall be designed an optimal microstructure to withstand abrasive wear?
13. Performance vs. properties: following questions n.1 and n.2, guess similar answers assuming a complex geometry and large size of the structural part
14. Why strain hardening ability is maximized in Cu and Cu-alloys?
15. Guess the degree of coherency of the precipitate interface at maximum peak strength upon correct aging?
16. What is the strengthening method enabling the maximum strength and toughness combination?
17. Which strengthening method does exert uniform plastic strain?
18. Assuming the presence of manufacturing defects in industrial parts, which mechanical property shall priority be used to meet safe design requirement?
19. Does the base metals in alloys affect their strength ?
20. What is the most effective dislocation source in metals?
21. Which are the primary factors in precipitation hardening?
22. Why substitutional atoms determine an increased strength in alloys?
23. Why of the upper and lower yield stress in some alloys?
24. Why of the Luders' bands during tensile test of some alloys

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Lecture 1

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Self-assessment Questions 3

11. Shall the angle of GB be small ($<15^\circ$) or large ($>15^\circ$) to enable strain hardening?
12. Is thermal expansion a microstructure-sensitive or insensitive property?
13. Why the first stage of aging is accompanied by a decrease in strength?
14. What is the most important performance which must be exhibited by castings in practical applications and how it is met in the manufacturing practice?
15. What is the simplest and most economical method to strengthen a metal?
16. In precipitation strengthening shall precipitates form inside the grains or at GBs?
17. Explain the phenomenon of Cottrell atmospheres and its consequences?
18. What is the common factor shared by all strengthening methods except that for fiber strengthening?
19. How toughening of castings is achieved using the eutectic modification?
20. What is the main strengthening method in HSLA steels?
21. What is the main strengthening method of aluminum alloy?
22. State the Schmid law in simple arguments?
23. Explain how the yield stress of an alloy changes upon cyclic strain hardening
24. How does the SFE affect the strain hardening of alloys
25. Which among BCC and FCC crystals is more ductile? Explain the reason.

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Lecture 1

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Self-assessment Questions 4

26. Explain why strengthening by solid solution is proportional to $\sqrt{\%C}$
27. Explain how ductility varies after solid solution strengthening of an alloy
28. Explain why certain alloy elements in certain alloys cause strengthening while other elements cause softening
29. Explain which lattice defects, other than dislocations, contribute to strain hardening and why
30. Clarify the importance of Orowan's equation on work hardening
31. Why are dislocation sources essential for strain hardening
32. Clear up in which conditions plastic deformation occurs by twinning rather than by slip
33. Explain the micro- and macroscopic phenomena of strain hardening with reference to both Ludwik-Hollomon and Taylor laws
34. Justify the large strain hardening coefficient of Co-alloys and Cu-alloys vs. that of cold rolled steels
35. Elucidate the role of SFE on dislocation arrangement upon ultrafast plastic deformation
36. Compare the consequences of strain aging and dynamic strain aging
37. Detail why strengthening by grain refinement enhance both strength and fatigue resist.

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Lecture 1

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